EEELunar Rover
Project Success Guide

A practical project-management, electronics, software, mechanical, testing, and report-writing guide for your ELEC40006 team

Project purpose
Design, build, test, and demonstrate a rover that can survey a lunar-style arena, identify each rock's age and type, manoeuvre quickly and accurately, and report discoveries through a user interface.

Key deadline	Date	What must be ready	Weighting
Interim presentation	28 May	15-minute presentation plus 5-minute Q&A showing progress, task allocation, project plan, and future timeline.	20%
Reflection + report	11 June	Group technical report as PDF, plus individual reflection forms attached at the end.	40% report; reflection must-pass
Interview, demo, competition	16 June	20-minute interview with functional rover, UI, circuit diagrams, code, test results, and 10-minute maximum demo run.	40%

Use this as the team's shared operating manual. Update it with your actual decisions, test data, and responsibilities as the project progresses.

Navigation Map

Use the numbered headings and PDF bookmarks to jump between sections. The most important day-to-day sections are the timeline, roles, test plan, and demo checklists.

Section	Use it for
1. Project objectives and success criteria	Make sure the team is solving the actual brief, not just building a moving robot.
2. System architecture	Understand how mechanical, electronics, software, sensing, and UI fit together.
3. Timeline and milestones	Track what must be done each week.
4. Delegation and team process	Assign ownership and keep workload fair.
5. Equipment and materials	Check what you need before lab sessions and orders.
6. Coding approach	Keep code modular, testable, and demo-safe.
7. Electronics approach	Build and debug sensor circuits systematically.
8. Mechanical approach	Keep the rover light, robust, and easy to repair.
9. Testing plan	Generate evidence and reduce final-demo risk.
10. Report and presentation plan	Protect marks by documenting as you build.
11. Demo and interview preparation	Prepare the rover and each teammate for final assessment.
12. Checklists and templates	Copy/paste practical templates into your team tracker.

1. Project objectives and success criteria

The rover must solve a complete engineering task: physical movement, sensing, classification, communication, reliability, and clear documentation. The project brief requires the rover to identify rock age using radio signals, identify rock type using infrared pulses, ultrasound, and magnetic-field direction, move quickly and accurately around the arena, and communicate with an app or webpage interface.

Objective	What success looks like	Evidence to collect
Reliable movement	Forward, reverse, left, right, stop, and speed control work repeatedly from the UI.	Short videos; movement calibration notes; operator test results.
Rock approach	Rover can be driven close enough to each rock for consistent sensing without heavy collisions.	Arena practice videos; notes on approach distance and orientation.
Rock age classification	Radio/age signal is detected, decoded, and displayed clearly.	Known-rock test table; raw readings; screenshots.
Rock type classification	IR, ultrasound, and magnetic readings combine into a repeatable type result.	Sensor calibration table; false-positive/false-negative notes.
User interface	Operator can drive, scan, and read final rock results quickly.	UI screenshots; command protocol; demo video.
Weight and robustness	Aim for under 750 g and secure all components for repeated runs.	Mass budget; final mass; photos of final assembly.
Assessment readiness	Team can explain system, circuits, code, tests, and design choices.	Interview notes; diagrams; test evidence folder.

Definition of success
The goal is not only to make the rover work once. The goal is a repeatable, explainable, documented rover that survives the final interview and demo.

2. System architecture

Treat the rover as connected subsystems. This prevents the common failure mode where individual parts work separately but cannot be integrated.

Subsystem	Purpose	Owner should produce
Mechanical platform	Carry components, keep sensors aligned, keep mass low, survive movement.	Final layout photos, mass budget, mounting notes.
Motor control	Drive accurately from user commands.	Motor functions, speed control, calibration notes.
Sensing and classification	Read rock properties and convert raw data into age/type.	Sensor circuits, raw readings, classification algorithm.
Communication and UI	Send commands, receive results, support quick demo operation.	Web/app interface, command protocol, screenshots.
Documentation and evidence	Turn design process into marks and interview confidence.	Report, test tables, diagrams, videos, risk register.

Recommended data flow:

Layer	Flow
User Interface	Operator sends movement/scan commands and sees

Layer	Flow
	status/results.
WiFi/communication	Commands and readings pass between UI and rover.
Microcontroller code	Receives commands, controls motors, reads sensors, runs classification.
Electronics	Motor driver, sensors, signal conditioning, power distribution.
Mechanical rover	Moves through arena and positions sensors near rocks.

3. Timeline and milestones

Project manager rule
Start integration early. Stop adding major new features before the final week. The final week should be for reliability, evidence, report completion, and rehearsal.

Period	Main goal	Technical outputs	Management/documentation outputs
Week 1: planning + first prototypes	Everyone knows what they own; rover can do something basic.	Kit checked; motors tested; WiFi tested; first UI command; sensor research started.	GitHub and shared drive created; roles assigned; Gantt chart draft; risk register started; report skeleton created.
Week 2: subsystem build	Each subsystem works alone before integration.	Improved driving; first IR/ultrasound/magnetic/radio tests; basic UI; temporary mechanical layout.	Interim slides; task allocation table; current progress evidence; future plan.
28 May: interim presentation	Show credible progress and a realistic finish plan.	Short demo/video if available; system architecture; early test results.	15-minute presentation plus Q&A; Gantt chart; risks and mitigations.
Week 3: integration	One system can drive, sense, classify, and report.	Merged code; final pin map; first full rock scan sequence; secure wiring begins.	Report 50-60% drafted; test tables populated; backup videos started.
Week 4: reliability + report	Freeze design and make it repeatable.	Full demo simulations; WiFi/battery/restart tests; final assembly.	Report finished; reflections attached; submission PDF checked; interview notes ready.
11 June: report submission	Submit polished report by 17:00.	Final design locked and documented.	PDF report submitted; GitHub shared if required.
16 June: interview/demo/competition	Explain and demonstrate with confidence.	Functional rover, UI, diagrams, code, test results, backup videos.	Each member ready to explain their subsystem; demo checklist complete.

4. Delegation and team process

Each area needs one clear owner, but nobody should work in isolation. The owner is responsible for progress, testing, and documentation of that area.

Role	Responsibilities	Key deliverables
Project Manager / Integration Lead	Own timeline, task board, risks, meetings, integration, fairness, and final readiness.	Gantt chart, risk register, decision log, integration checklist.
Electronics Lead	Sensor circuits, power distribution, pin map, oscilloscope testing, wiring quality.	Schematics, oscilloscope screenshots, sensor test evidence.
Embedded Software Lead	Microcontroller code, motor functions, sensor functions, classification, GitHub	Clean code, README, flowchart,

Role	Responsibilities	Key deliverables
	code hygiene.	classification explanation.
UI / Communications Lead	Web/app interface, WiFi communication, command protocol, result display.	Working UI, screenshots, protocol table, connection test results.
Mechanical / Integration Engineer	Chassis layout, mounting, mass budget, cable management, physical robustness.	Mechanical photos/diagrams, final mass, assembly checklist.
Documentation / Testing Lead	Report structure, test plan, evidence collection, presentation, reflection coordination.	Report draft, test tables, slides, final submission checklist.

Meeting type	Frequency	Agenda
10-minute stand-up	Every working session	What was finished? What is next? What is blocked? Who needs help?
Integration review	Twice per week	Does the rover still drive? Do sensors still work? What changed? What evidence was captured?
End-of-week review	Weekly	Update Gantt chart, risk register, report, test results, and next priorities.

5. Equipment and materials

Category	Items
Core kit	Rover platform; Adafruit Metro M0 with WiFi shield; motor driver module; rock simulator; provided batteries.
Electronics	Breadboard/prototyping board, jumper wires, resistors, capacitors, IR components, ultrasound parts, magnetic sensor, radio/age detection components, switches/buttons, LEDs, headers/connectors.
Lab equipment	Oscilloscope, multimeter, bench power supply for safe circuit testing, Picoscope if borrowed, scales, laptop/toolchain, camera/phone for evidence.
Mechanical materials	Lightweight brackets/plates, Velcro, cable ties, double-sided foam tape, screws, spacers, labels.
Documentation tools	GitHub repository, shared drive, task board, report document, test spreadsheet, photo/video evidence folder.

Battery safety
Only use the approved batteries. Do not recharge, short, drop, damage, or misuse batteries. Check polarity before insertion and keep terminals away from conductive materials.

6. Coding approach

Use simple, modular, testable code. The code should be understandable by more than one teammate and easy to demonstrate during the interview.

Coding principle	Implementation
Modular structure	Separate motor control, WiFi/server communication, IR sensing, ultrasound sensing, magnetic sensing, radio/age sensing,

Coding principle	Implementation
	classification, and logging.
State machine	Use states such as IDLE, DRIVING_FORWARD, TURNING_LEFT, SCANNING_ROCK, CLASSIFYING, REPORTING_RESULT, and ERROR.
Test modes	Add commands for motor test, IR-only read, ultrasound-only read, magnetic-only read, radio-only read, and full scan.
Serial logging	Print raw sensor values, classification decisions, WiFi status, and error messages. Save useful logs for the report.
Communication protocol	Use clear commands such as FWD, BACK, LEFT, RIGHT, STOP, SPEED:120, SCAN, READ_IR, READ_US, READ_MAG, READ_RADIO.
Version control	Commit stable versions frequently. Keep a known-good demo version. Use meaningful commit messages.

Suggested repository folder	Contents
/code	Microcontroller, UI, helper scripts.
/circuit_diagrams	Schematics, pin maps, oscilloscope notes.
/mechanical_design	Chassis photos, mount sketches, mass budget.
/test_results	Sensor readings, calibration tables, demo run timings.
/report	Report draft, figures, references, reflection forms.
/presentation	Interim slides and speaker notes.
/README.md	Setup instructions, dependencies, known-good version notes.

7. Electronics approach

Build and debug electronics one subsystem at a time. Do not wire everything at once.

1. Understand each signal: expected voltage, current, timing, frequency, analogue/digital behaviour, range, and orientation.
2. Prototype one sensor at a time using known rock settings.
3. Record raw readings before writing final classification rules.
4. Add signal conditioning only when data shows it is needed: current limiting, pull-ups/pull-downs, voltage dividers, amplification, filtering, or decoupling.
5. Test sensors while motors are running to detect electrical noise or voltage dips.
6. Prepare oscilloscope probe points and expected waveforms for the interview.

Electronics risk	Typical cause	Mitigation
Noisy sensor values	Motor noise, long wires, weak signal, poor grounding.	Common ground, shorter wires, decoupling, filtering, stop-before-scan behaviour.
Sensor saturation	Too close/far from rock or poor resistor values.	Calibrate range; adjust mounting; change component values.
WiFi or MCU reset	Voltage dip during motor start.	Check supply; reduce motor load; add decoupling; test battery condition.
Loose connections	Breadboard vibration and movement.	Secure wires, label connections, consider prototyping board for final layout.

8. Mechanical approach

Mechanical priority	Practical action
Weight	Aim below 750 g. Start a mass budget in Week 1 and update it after every addition.
Sensor placement	Mount sensors at consistent height and angle; keep them unobstructed; make early mounts adjustable.
Balance	Keep heavy parts low and central; prevent tipping during turns.
Access	Leave access to battery, USB, reset button, LEDs, and oscilloscope probe points.
Cable management	Keep wires away from wheels; use ties/tape; label important connections.
Arena performance	Test on the actual or similar surface; check grip, turning radius, and approach distance.

Mechanical design stages
Stage 1: temporary layout for fast testing. Stage 2: semi-fixed layout once sensor positions are known. Stage 3: final demo layout with secure wiring, labels, and tested assembly.

9. Testing plan

Testing is the main way to turn a fragile prototype into a reliable engineering system. Every subsystem should have a pass/fail condition and evidence.

Subsystem	Test	Pass condition
Motors	Forward/back/turn/stop from UI.	Correct response every time; emergency stop works.
WiFi	Connect and send repeated commands for 10 minutes.	No random disconnects or unsafe behaviour.
IR sensing	Read known rock settings repeatedly.	Values are distinguishable and repeatable.
Ultrasound	Read known distance/signal conditions.	Stable readings within acceptable tolerance.
Magnetic sensing	Detect known directions/configurations.	Correct direction/classification across repeats.
Radio/age	Decode known age signal.	Correct age displayed consistently.
UI	Drive, scan, display, and log result.	Operator can use it under time pressure.
Full system	Drive to rock, scan, classify, record result, repeat.	Complete demo-style run under target time with accurate results.

Evidence type	What to capture
Photos	Circuit, mechanical layout, sensor mounting, final rover.
Videos	Movement, successful scan, full arena-style run, fallback proof.
Screenshots	UI, serial monitor, GitHub history, oscilloscope captures.
Tables	Calibration readings, repeated trials, classification accuracy, demo timings.
Notes	Failure causes, design decisions, fixes, limitations.

10. Report and presentation plan

The report is a major deliverable, so write it while building. Do not wait until the rover is finished.

Report section	What to include
Cover page	Module, project title, group number, all names, CIDs, word count, GitHub link if required.
Executive summary	What was built, final capability, main design choices, final performance.
Introduction and objectives	Challenge context, functional requirements, non-functional requirements.
Requirements/specification	Movement, sensing, UI, weight, reliability, cost, safety.
Project management	Team structure, timeline, Gantt chart, task allocation, risk register, decision process.
System design	Block diagram, subsystem explanations, data/power flow.
Mechanical design	Chassis, sensor mounting, mass budget, final assembly.
Electronics design	Schematics, power, sensor circuits, signal conditioning, oscilloscope evidence.
Software design	Architecture, state machine, protocol, UI, classification algorithm, error handling.
Testing and results	Unit tests, integration tests, calibration, accuracy, timings, failures and fixes.
Discussion	Trade-offs, what worked, what failed, limitations, future improvements.
Safety/professional practice	Battery safety, teamwork, EDI, communication, self-learning, sustainability.
Conclusion	Final outcome against objectives.
Appendices	Full data, extra diagrams, key code excerpts, risk register, individual reflections.

Interim presentation slide	Purpose
1. Title and roles	Show team organisation.
2. Challenge and requirements	Show understanding of the brief.
3. System architecture	Explain complete design.
4. Mechanical design	Show layout and weight thinking.
5. Electronics/sensors	Show circuits and sensing plan.
6. Software/UI	Show command and classification approach.
7. Current progress/demo	Show working evidence.
8. Test results so far	Show engineering data.
9. Timeline/Gantt chart	Show finish plan.
10. Risks and next steps	Show control of uncertainty.

11. Demo and interview preparation

Team member	Must be ready to explain
Project Manager	Overall system, timeline, risk management, integration, final trade-offs.
Electronics Lead	Circuits, sensor signals, power, oscilloscope probing, failure

Team member	Must be ready to explain
	modes.
Software Lead	Code structure, state machine, classification logic, debugging process.
UI/Comms Lead	Communication protocol, interface operation, WiFi fallback.
Mechanical Lead	Chassis, sensor mounting, mass control, robustness.
Testing/Report Lead	Test plan, results, evidence, limitations, report structure.

Demo-day checklist	Done?
Rover code uploaded and known-good version tagged.	
Laptop charged and UI tested.	
Battery checked; polarity checked; no loose shorts.	
Emergency stop tested.	
Motors tested briefly.	
One sensor reading tested.	
Circuit diagram, pin map, test results, and code ready to show.	
Backup videos available.	
Group number label on chassis.	
Everyone knows their explanation.	

12. Checklists and templates

12.1 Power-on checklist

Step	Check
1	Inspect wires for loose ends or shorts.
2	Check battery polarity and approved battery type.
3	Confirm wheels are free and not blocked by wires.
4	Power on rover.
5	Confirm microcontroller boots.
6	Connect UI.
7	Test stop command first.
8	Test motors briefly.
9	Run one sensor read.
10	Begin demo/test only if all checks pass.

12.2 Risk register template

Risk	Likelihood	Impact	Prevention	Backup
Sensor readings unreliable	High	High	Test early with known configurations.	Repeat scan; display confidence/warnings.
WiFi fails in demo	Medium	High	Test local network and	Use phone hotspot or

Risk	Likelihood	Impact	Prevention	Backup
			hotspot.	local router.
Rover too heavy	Medium	Medium	Mass budget from Week 1.	Remove non-essential parts.
Motor noise affects sensors	Medium	High	Test sensors while motors run.	Stop before scanning; add filtering.
Code breaks late	Medium	High	Use GitHub and stable tags.	Revert to known-good commit.
Report rushed	High	High	Start report Week 1.	Use documentation lead and templates.

12.3 Sensor test table template

Rock setting	IR reading	Ultrasound result	Magnetic direction	Radio/age result	Expected type	Rover prediction	Pass?
A							
B							
C							
D							

12.4 Definition of done

Subsystem	Done means
Motor control	Works from UI, stops safely, has calibration notes, tested after reboot.
Sensor circuit	Reads known configurations, has schematic, raw data table, and mounting notes.
Classification	Rules are documented, tested on repeated trials, and limitations are known.
UI	A teammate can drive and scan without developer help; screenshots captured.
Mechanical assembly	Everything secured, labelled, under mass target if possible, and tested after movement.
Report section	Written, evidence included, reviewed by another teammate, figures labelled.

13. Final operating rules for success

7. A working simple rover beats an ambitious broken rover.
8. Integrate early and often; no subsystem should stay isolated for more than a few days.
9. Freeze major design changes before the final week.
10. Record evidence whenever something works.
11. Every subsystem needs a named owner and a test result.
12. At least two people must know how to upload code, run the UI, and operate the rover.
13. Keep the report alive from day one.
14. Be honest about failures and show how you fixed or managed them.

Source notes

This guide is based on the ELEC40006 Deliverables Guidance and the ELEC40006 Rover to the Moon project briefing supplied to the team. It summarises the deadlines, assessment requirements, technical requirements, safety notes, and project advice from those documents into a practical team guide.

Course documents referenced:

- ELEC40006 Deliverables Guidance 2526.
- ELEC40006 Electronics Engineering Design Project 1 2526 - Rover to the Moon!